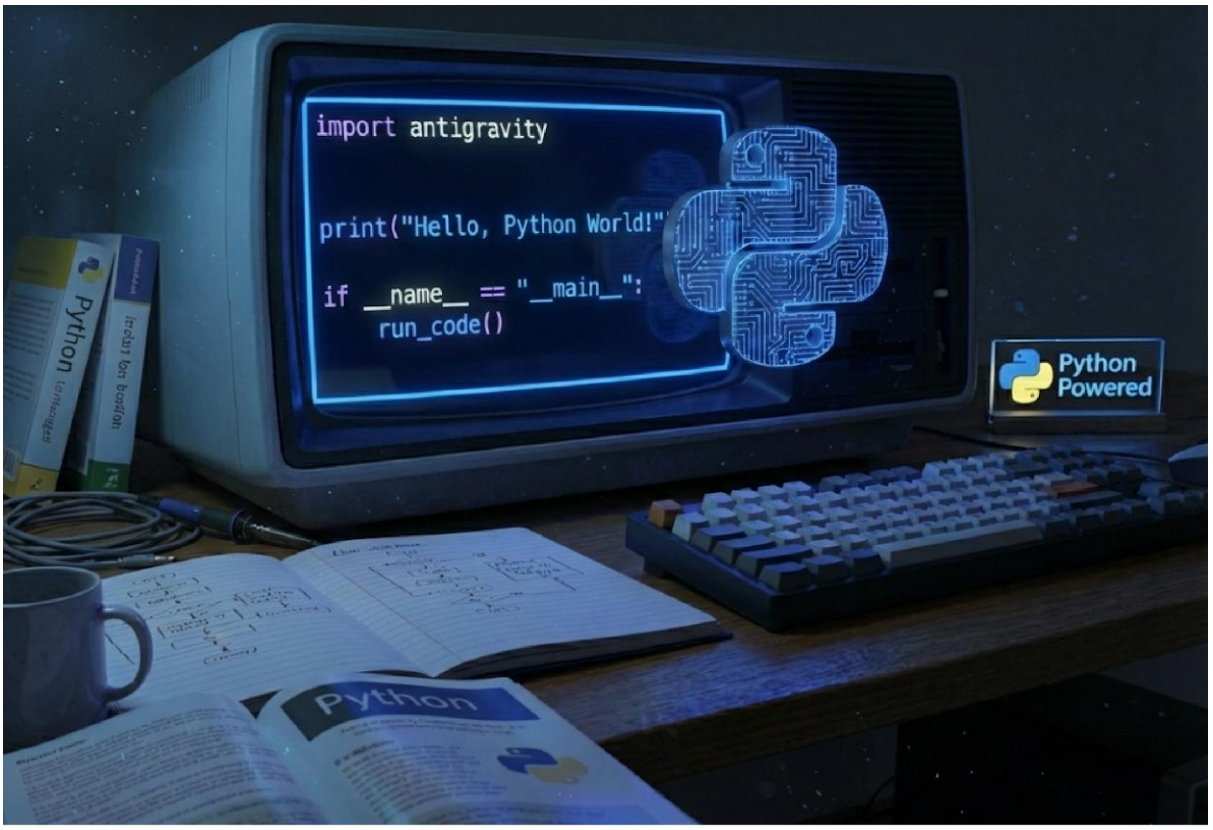
MIT | **Academy of Engineering**

(An Autonomous Institute Affiliated to Savitribai Phule Pune University)

Subject (Code):
Essentials of Data Science
(2304102L)

EDS CodeTantra Programs

Name	(PRN)	Div- Batch
TANMAY Rahul MUTKULE	(202501090093)	ME2 – M21



Practical 01

1.1. Practice Lab Assignments

1.1.1. Calculate Momentum

Write a program that accepts the mass of an object (in kilograms) and its velocity (in meters per second), then calculates and displays the momentum of the object. The momentum p is calculated using the formula: $p = m v$

where:

p is the mass of the object (in kilograms).

m is the velocity of the object (in meters per second).

Input Format:

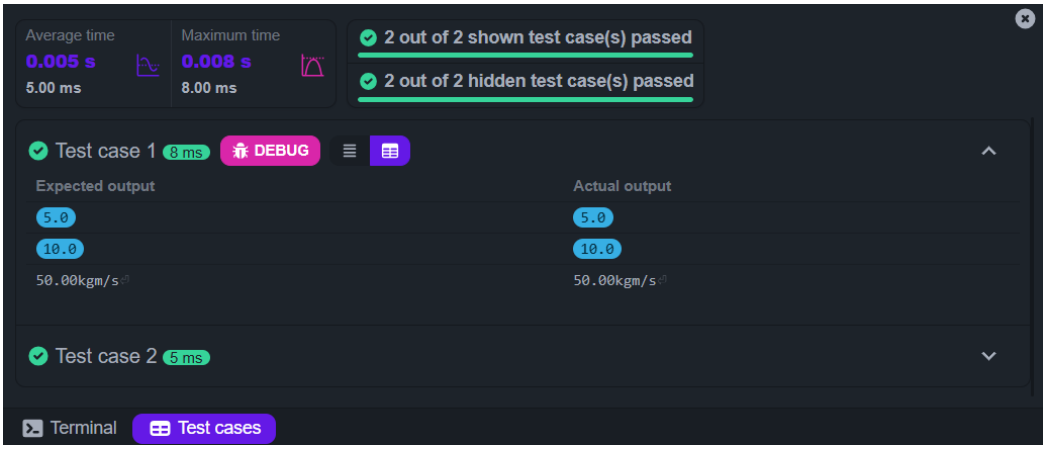
A single floating-point number representing the mass of the object in kilograms.

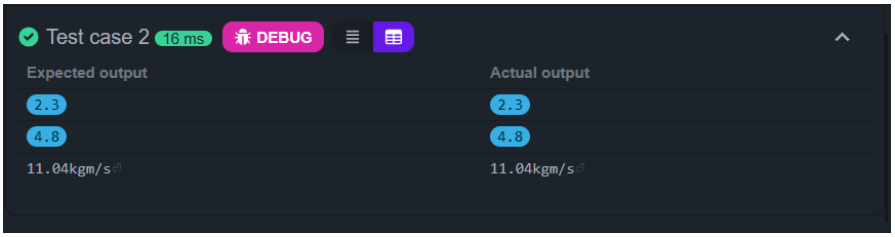
A single floating-point number representing the velocity of the object in meters per second.

Output Format:

The output will display calculated momentum with appropriate units (kgm/s) (rounded up to 2 decimal places).

```
mass=float(input())
velocity=float(input())
momentum=mass*velocity
print(f"{momentum:.2f}kgm/s")
```





1.1.2 Conditional Calculation based on the Number of Digits

Write a Python program that accepts an integer as input. Depending on the number of digits in *n*.

Constraints:

$1 \leq n \leq 999$

Input Format:

The input consists of a single integer *n*.

Output Format:

If *n* is a single-digit number, print its square.

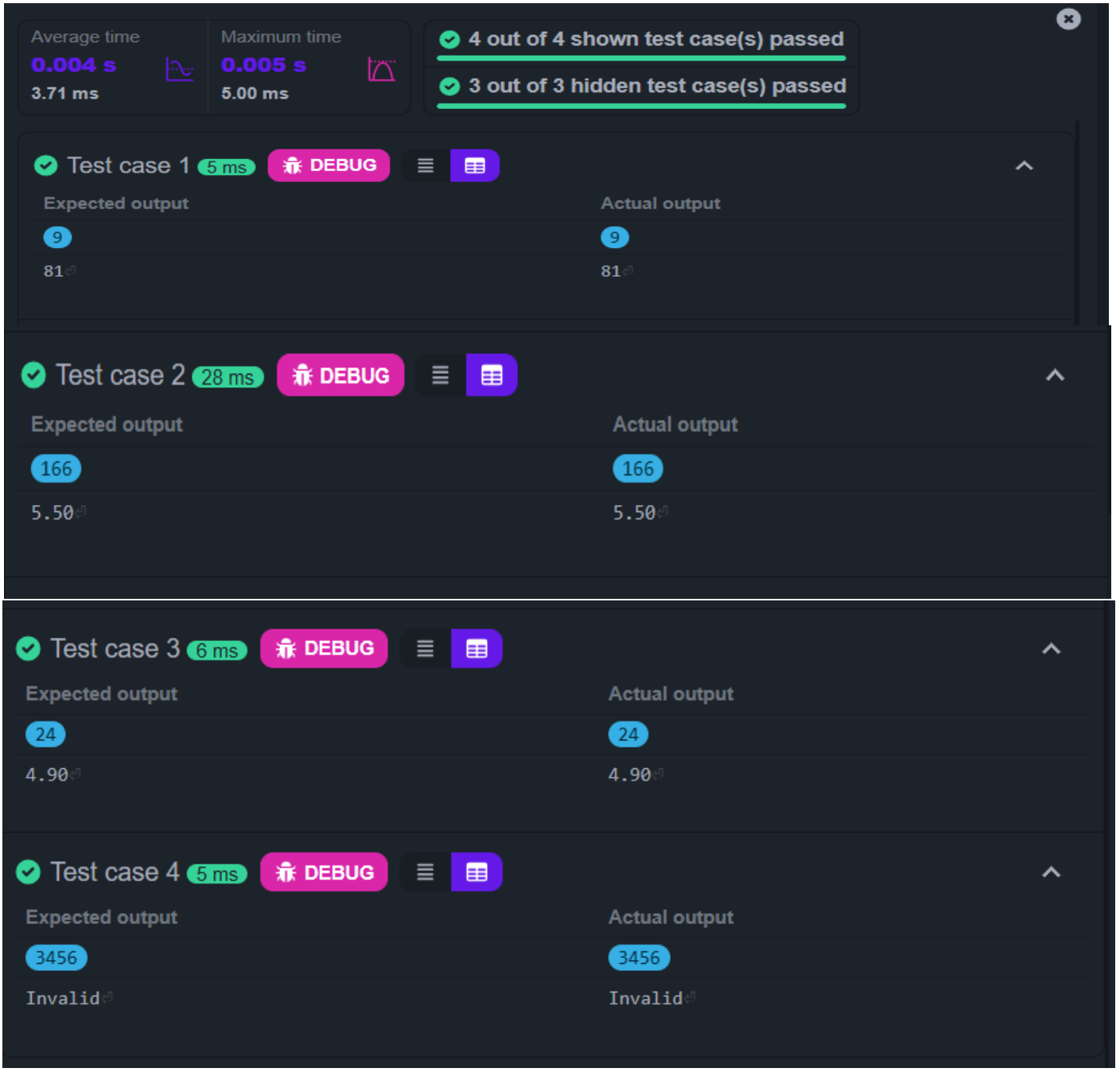
If *n* is a two-digit number, print its square root (rounded to two decimal places).

If *n* is a three-digit number, print its cube root (rounded to two decimal places).

Else print "Invalid".

```
import math

number=int(input())

if 0<= number <=9:

        print(number**2)

elif 10<= number <=99:

        print(f"{math.sqrt(number):.2f}")
```

```
elif 100<= number <=999:
        print(f"{math.cbrt(number):.2f}")
else:
        print("Invalid")
```



1.1.3. Days Between Two Dates

Write a Python program to calculate the number of days between two dates.

Input Format:

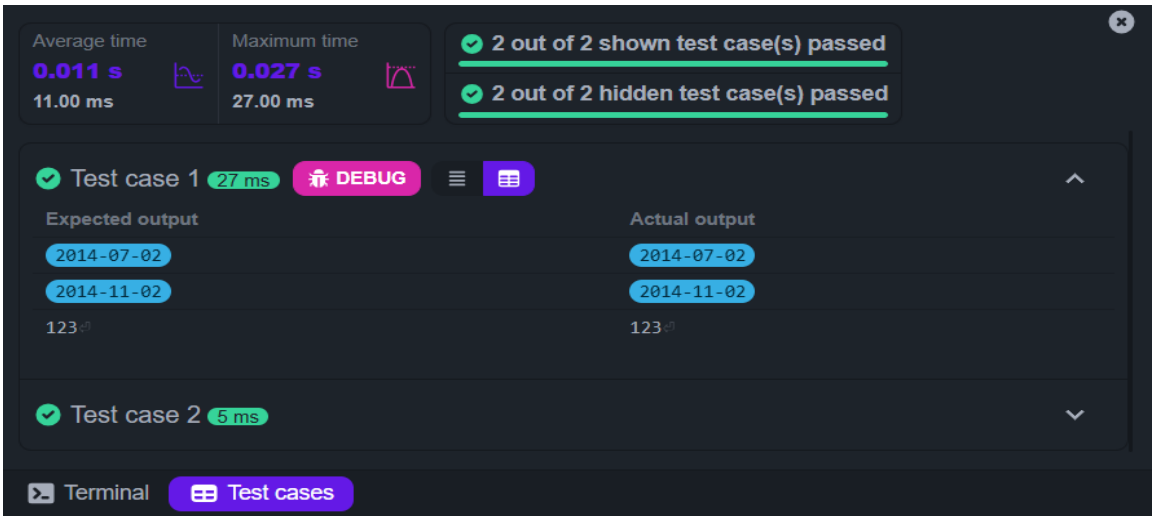
The first line contains the first date in the format *YYYY–MM–DD*.

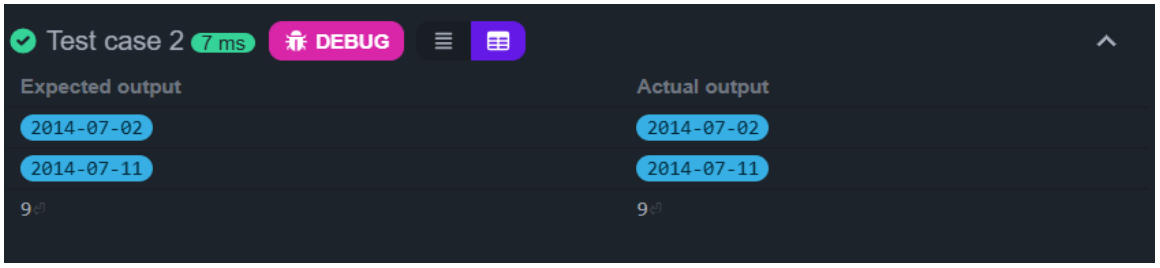
The second line contains the second date in the format *YYYY–MM–DD*.

Output Format:

An integer representing the number of days between the given dates.

```
from datetime import datetime


date1=input().strip()
date2=input().strip()


d1=datetime.strptime(date1, "%Y-%m-%d")
d2=datetime.strptime(date2, "%Y-%m-%d")
difference=d2-d1
print(difference.days)
```





1.1.4. Reverse a Number

Write a Python program to reverse the digits of the number and print the reversed number.

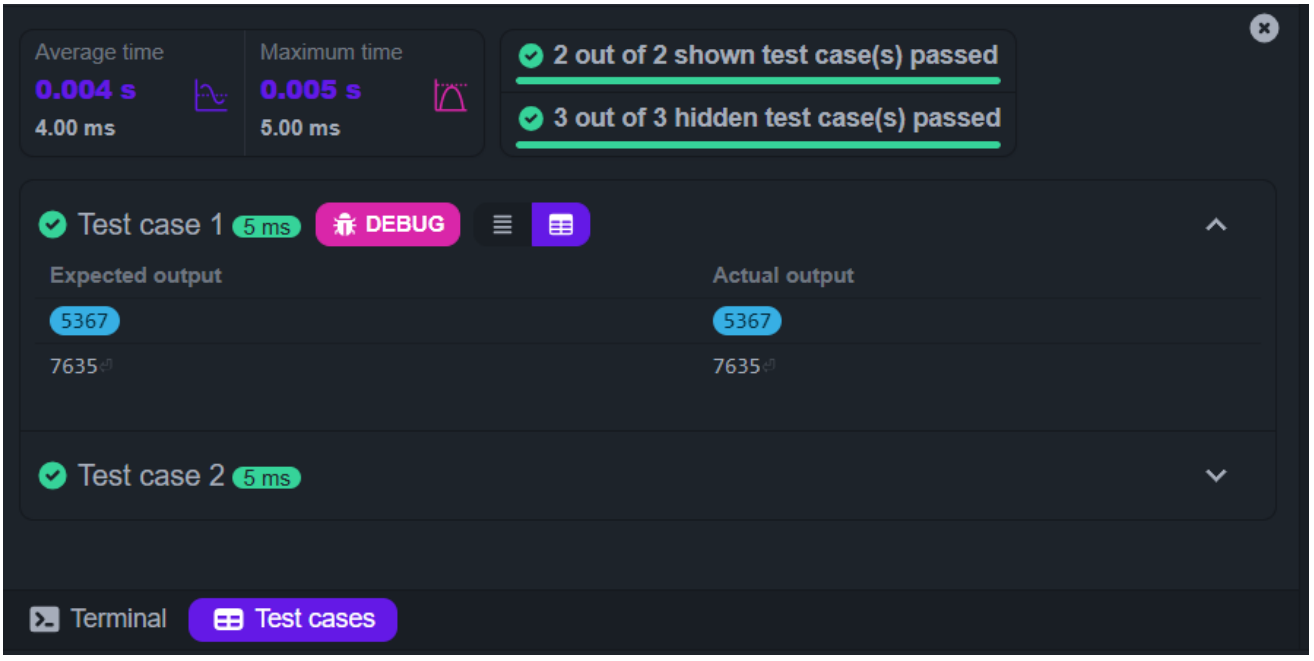
Input Format:

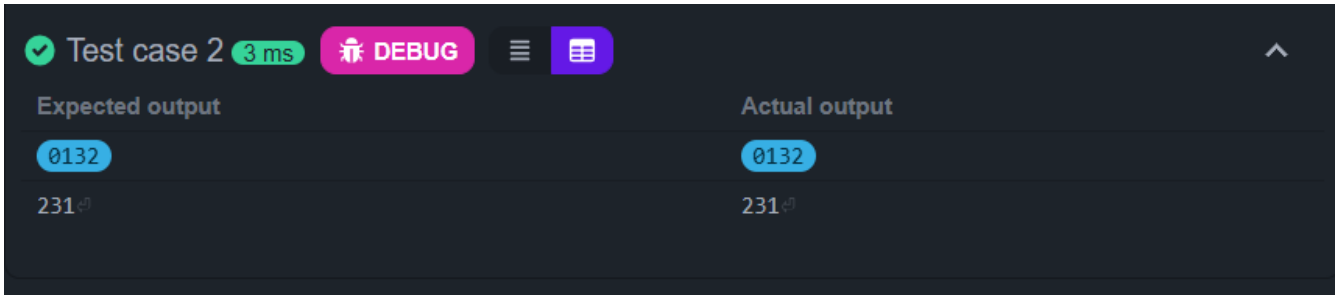
- The input is a single integer.

Output Format:

- The output prints a single integer, which is the reversed number.

```
n = int(input())

reversed_num=int(str(num)[::-1])

print(reversed_num)
```





1.1.5. Multiplication Table

Write a Python program that takes an integer as input and prints the multiplication table for that integer from 1 to 10.

Input Format:

- The first line of input contains an integer that represents the number for which the multiplication table is to be printed.

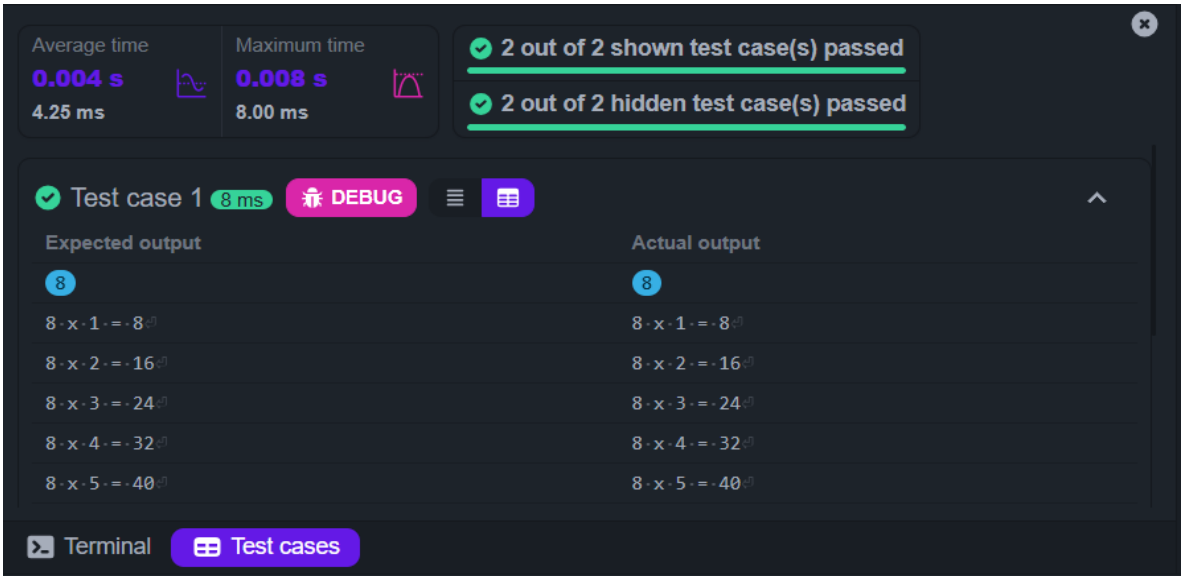
Output Format:

- Print the multiplication table for the given number.
  <number> x <multiplier> = <result>

Note:

- The character "x" is used in the output to represent multiplication.
- Refer to the visible test-cases for more clarity.

```
num = int(input())

for i in range(1, 11):
    print(f"{num} x {i} = {num * i}")
```



Test case 2 4 ms DEBUG

Expected output	Actual output
5	5
5 x 1 = 5	5 x 1 = 5
5 x 2 = 10	5 x 2 = 10
5 x 3 = 15	5 x 3 = 15
5 x 4 = 20	5 x 4 = 20
5 x 5 = 25	5 x 5 = 25
5 x 6 = 30	5 x 6 = 30
5 x 7 = 35	5 x 7 = 35
5 x 8 = 40	5 x 8 = 40
5 x 9 = 45	5 x 9 = 45
5 x 10 = 50	5 x 10 = 50

1.2 Lab Assignments

1.2.1. Pass or Fail

Write a Python program that accepts the number of courses and the marks of a student in those courses.

The grade is determined based on the aggregate percentage:

- If the aggregate percentage is greater than 75, the grade is Distinction.
- If the aggregate percentage is greater than or equal to 60 but less than 75, the grade is First Division.
- If the aggregate percentage is greater than or equal to 50 but less than 60, the grade is Second Division.
- If the aggregate percentage is greater than or equal to 40 but less than 50, the grade is Third Division.

Input Format:

- The first input will be an integer *n*, the number of courses.
- The second input will be *n* integers representing the marks of the student in each of the *n* courses, separated by a space.

Output Format:

If the student passes all courses:

- Print the aggregate percentage (formatted to two decimal places).
- Print the grade based on the aggregate percentage.

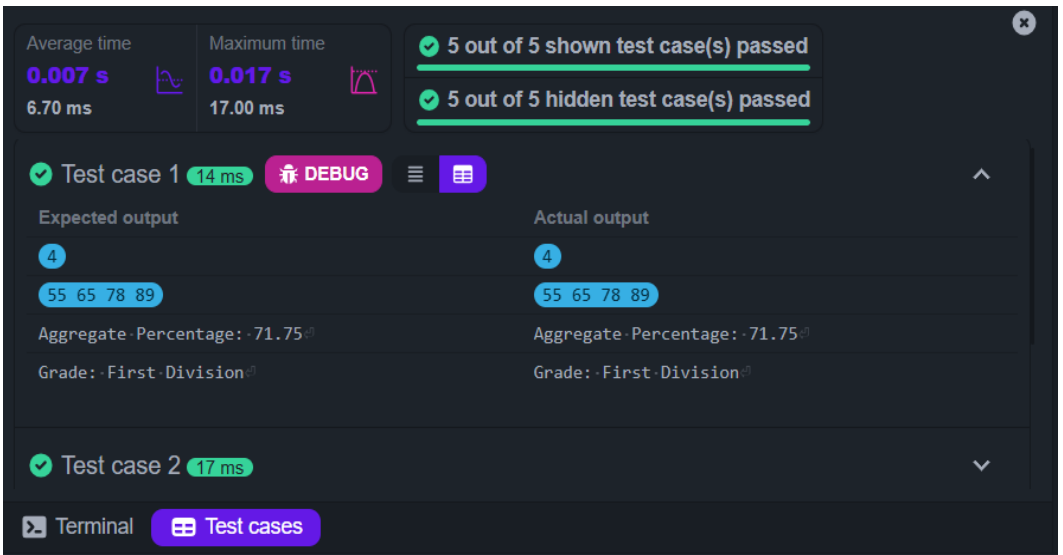
If the student fails any course (marks < 40 in any course), print:

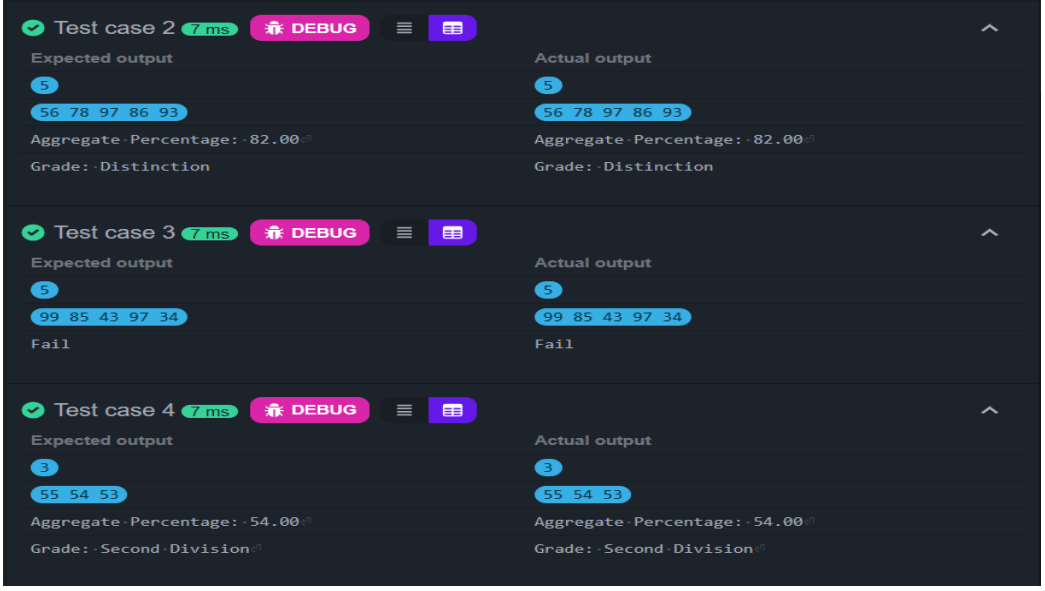
- "Fail".

Note:

- Aggregate Percentage: Average performance across all courses, calculated by summing all marks and dividing by the number of courses.

```
n = int(input())
marks= list(map(int,input().split()))
failed= False
for m in marks:

    if m<40:
        failed=True
        break

if failed:
    print("Fail")
else:
    total= sum(marks)
    agg=total/n
    print("Aggregate Percentage: "f"{agg:.2f}")

    if agg > 75:
        print("Grade: Distinction")
    elif agg >=60:
        print("Grade: First Division")
    elif agg>=50:
        print("Grade: Second Division")
    elif agg >=40:
        print("Grade: Third Division")
```





1.2.2. Fibonacci Series using Recursive Function

Write a Python program that uses recursion to print the first n terms of the Fibonacci series.

Input Format:

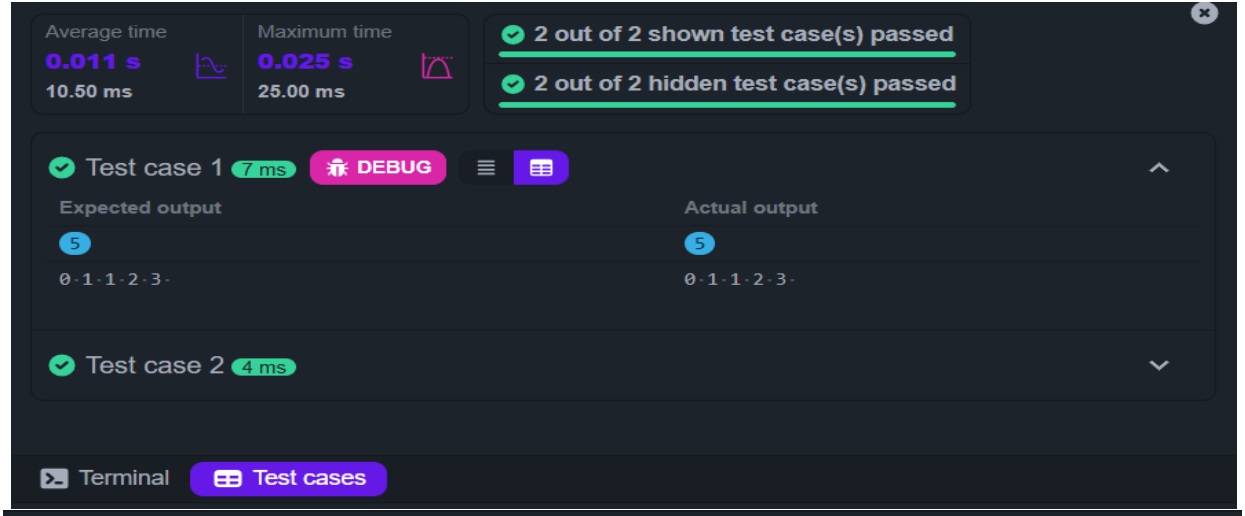
A single integer n representing the number of terms of generate.

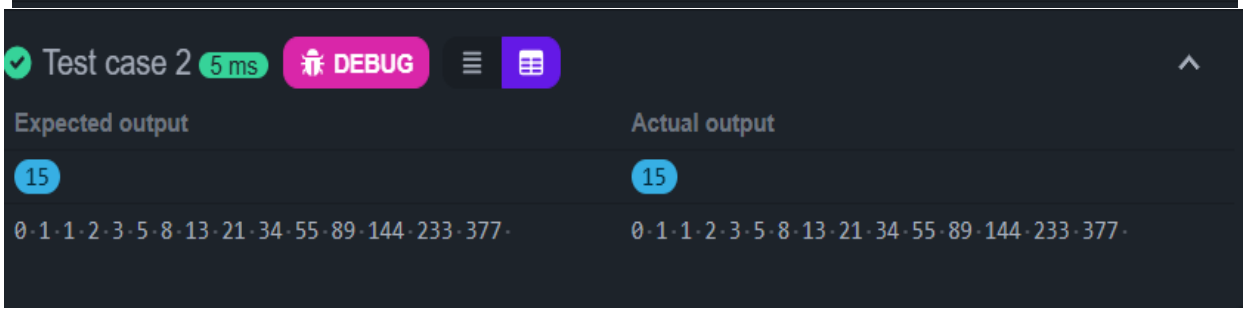
Output Format:

A single line containing the first n terms of the Fibonacci sequence, separated by spaces.

```
def fibonacci(n):
        if n==1:
            return 0
        elif n==2:
            return 1
        else:
            return fibonacci(n-1) + fibonacci(n-2)
```

```
n = int(input())
for i in range(1, n + 1):
    print(fibonacci(i), end=" ")
```





1.2.3. Pattern-1

Write a Python program to print a pattern of asterisks in the form of a right-angled triangle.

Input Format:

- The input is an integer, representing the number of rows in the pattern.

Output Format

- The output should display the pattern of asterisks (*), with each row containing an increasing number of asterisks.

Note: Refer to the displayed test cases for the sample pattern.

Test case 1:

4

*.⏎

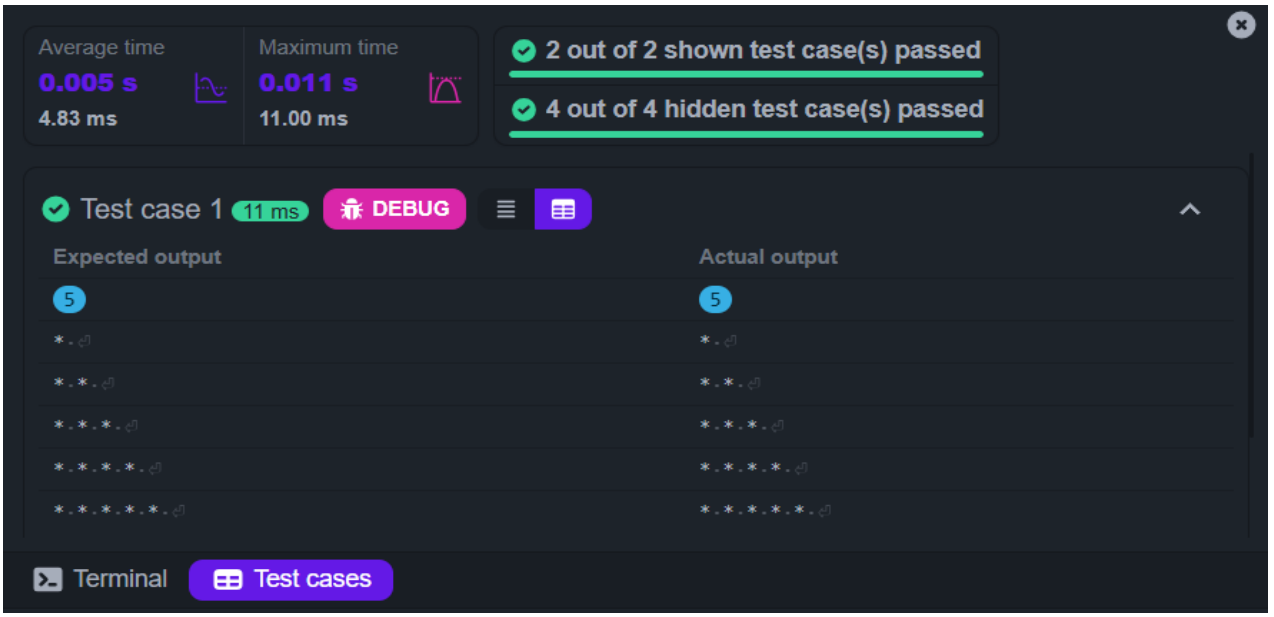
..⏎

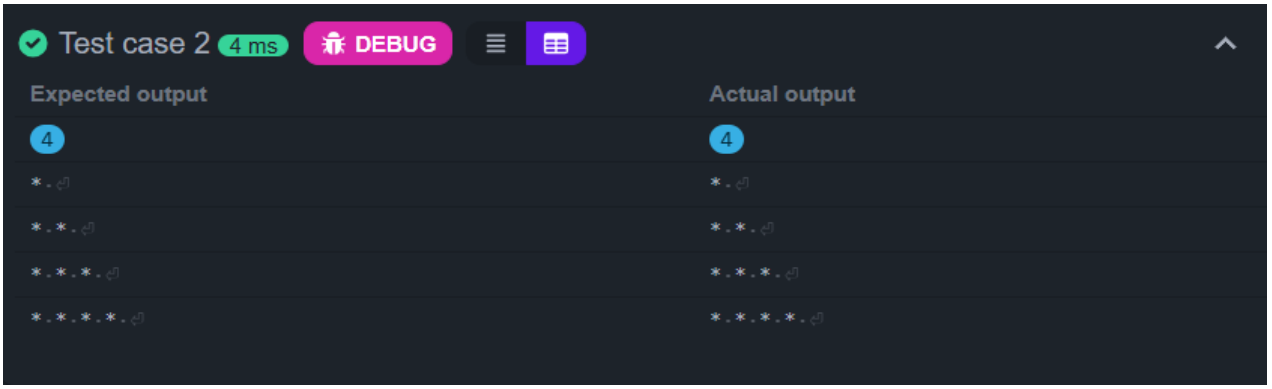
..*.⏎

..*.*.⏎

```
n = int(input())
for i in range(1, n + 1):
```

```
print('* '    * i)
```





1.

2.3. Pattern-2

Write a Python program to print a right-angled triangle pattern of numbers.

Input Format:

- The input is an integer, representing the number of rows in the pattern.

Output Format:

- The output should display the pattern of numbers separated by space, with each row containing increasing numbers starting from 1 up to the row number.

Note:

- Refer to the displayed test cases for the sample pattern.

Test case 1:

5

1· ⏎

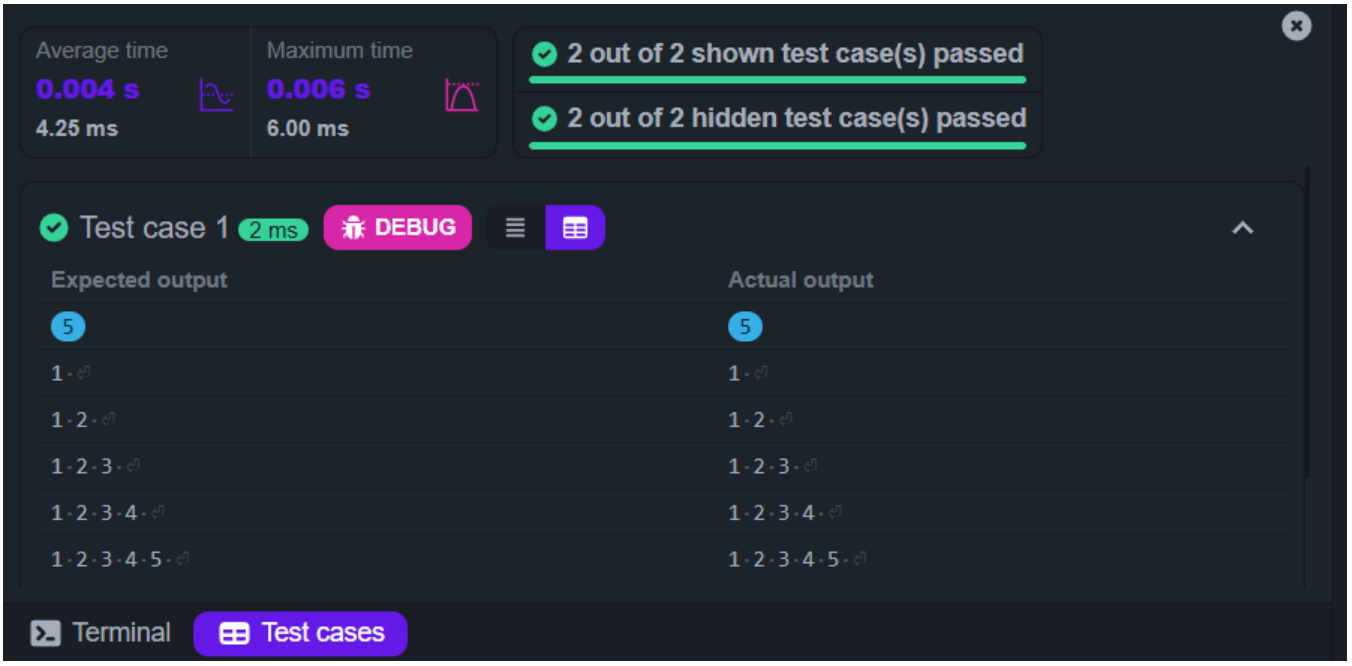
1· 2· ⏎

1· 2· 3· ⏎

1· 2· 3· 4· ⏎

1· 2 · 3 · 4 · 5 · ⏎

```
n = int(input())
for i in range(1, 1+n):
    for j in range(1, 1+i):
        print(j, end=" ")
    print("")
```



Test case 2 5 ms DEBUG

Expected output	Actual output
8	8
1	1
1 2	1 2
1 2 3	1 2 3
1 2 3 4	1 2 3 4
1 2 3 4 5	1 2 3 4 5
1 2 3 4 5 6	1 2 3 4 5 6
1 2 3 4 5 6 7	1 2 3 4 5 6 7
1 2 3 4 5 6 7 8	1 2 3 4 5 6 7 8